

Trainings ▾

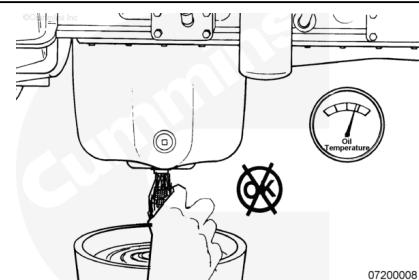
Drain

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Operate the engine until the water temperature reaches 60°C [140°F].

Shut off the engine.

Remove the oil drain plug.

Drain the oil immediately to make sure all the oil and suspended contaminants are removed from the engine.

If lubricating oil element failure occurs resulting in contamination of the lubricating oil, the following is recommended:

- Inspect the lubricating oil filter by cutting it open and inspecting for debris. Refer to Procedure 007-083 (</qs3/pubsys2/xml/en/procedures/10/10-007-083.html>).
- If debris is **not** found, drain the lubricating oil and remove the lubricating oil pan. Clean and reinstall the lubricating oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>).
- Remove the valve cover and clean the contaminated lubricating oil from the cylinder head cavities. This can be done using a siphoning device. Refer to Procedure 003-

011 (/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html).

- If debris is found, refer to the Cummins Warranty Administration Manual, Bulletin 4021290.

Install the oil pan drain plug(s) before adding engine oil to the system.

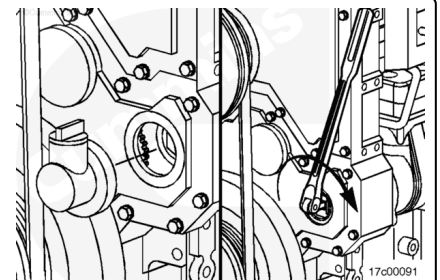
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Torque Value: 55 n•m [41 ft-lb]

Prime

⚠ CAUTION ⚠

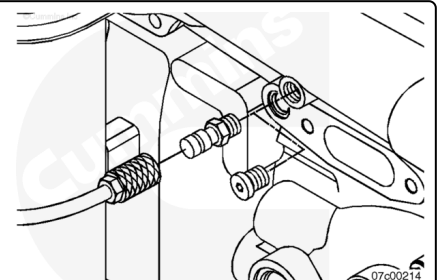
The lubricating oil system must be primed before operating the engine after any internal engine repairs or extended engine storage (beyond 6 months) to avoid internal component damage.



Bar the engine during oil priming.

Remove the oil plug from the lubricating oil cooler housing and install fitting and hose.

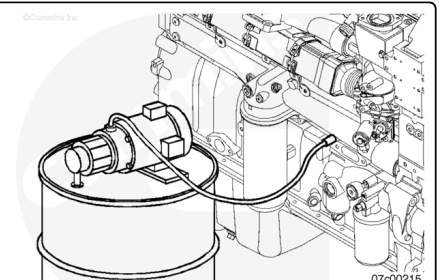
Use the coupler, Part Number 3376859, to connect the priming pump to the coupling.



Connect the priming pump oil supply hose to the lubricating oil pump coupling. Use clean 15W-40 lubricating oil from a drum or a container to supply oil to the lubricating oil pump and engine.

Turn the priming pump on.

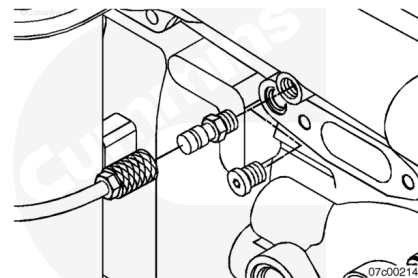
Allow the oil to flow until the oil pressure gauge indicates a maximum pressure of 69 to 138 kPa [10 to 20 psi] at the main oil rifle.



Turn the priming pump off.

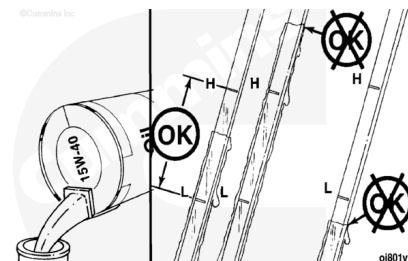
Remove the priming pump oil supply hose and fitting. Install the oil cooler housing plug.

Torque Value: 34 n•m [25 ft-lb]



Check the oil level on the dipstick before adding additional oil.

If necessary, fill the engine with lubricating oil.

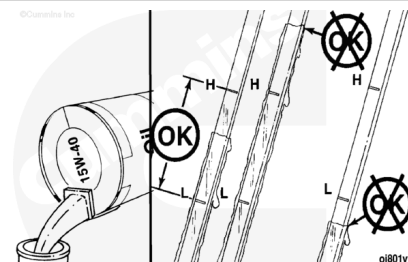


Fill

Fill and install the lubricating oil filter before putting oil into the engine. Refer to Procedure 007-013

(/qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html).

Add the specified amount of clean 15W-40 oil to fill the oil pan.



Oil Pan Capacity

Automotive and Industrial

45.42 liters [12 gal]

Power Generation

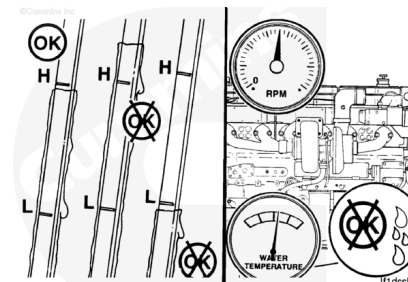
83.3 liters [22 gal]

Check the oil level on the dipstick. It **must** be filled to the H (high) mark.

Refer to Procedure 007-009

(/qs3/pubsys2/xml/en/procedures/10/10-007-009.html) for dipstick calibration.

Operate the engine until the water temperature reaches 82°C [180°F], and check for leaks.



Shut off the engine. Wait 15 minutes for the oil to drain back to the oil pan.

Check the oil level. Add oil as necessary to bring the level up to the H (high) mark on the dipstick.

